

# Are Points in Tennis Independent and Identically Distributed? Evidence From a Dynamic Binary Panel Data Model\*

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Forthcoming in the  
*Journal of the American Statistical Association*, 96,  
June 2001

## Abstract

This article tests whether points in tennis are independent and/or identically distributed (iid). We model the probability of winning a point on service and show that points are neither independent nor identically distributed: winning the previous point has a positive effect on winning the current point, and at ‘important’ points it is more difficult for the server to win the point than at less important points. Furthermore, the weaker a player, the stronger are these effects. Deviations from iid are, however, small and hence the iid hypothesis will still provide a good approximation in many cases. The results are based on a large panel of matches played at Wimbledon 1992–1995, in total almost 90,000 points. Our panel data model takes account of the binary character of the dependent variable, uses random effects to capture the unobserved part of a player’s quality, and includes dynamic explanatory variables.

**Keywords:** Dependence, non-identical distribution, tennis, binary choice, dynamic panel data, linear probability model, random effects.

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# 1 Introduction

This paper attempts to model the probability of winning a point on service in professional tennis. This probability is the key to most analyses in tennis. For instance, it provides the basis for the calculation of the probability of winning a match. In most papers on tennis, points are assumed to be independent and identically distributed (iid), which implies that the key probability is constant for a player throughout a match.

In this paper we test the iid hypothesis and reject it. We present a model that explicitly captures both the dependence (measured by the impact of the previous point) and the non-identical distribution (measured by the ‘importance’ of the current point) and use it to analyze the deviations from iid in more detail. We find that weaker players violate the iid hypothesis more than stronger players. Deviations from the iid hypothesis, although statistically strongly significant, are not large, and therefore the hypothesis will serve as a reasonable first-order approximation in many applications.

As a preview of our results (fully reported in section 4.3), let us consider a match in the men’s singles (women’s singles) between two average players. Overall these players will win 65% (56%) of their service points. If the previous point was won (and if the current point is not the first point in the game), then the probability of winning a point increases by 0.3% (0.5%), reflecting a ‘winning mood’. However, if the previous point was lost, then the probability of winning a point decreases by 0.5% (0.7%). Also, the more ‘important’ (defined in section 4.2) is the point, the lower is the probability that server wins the point. For example, at a point of zero importance, the probability of winning a point on service increases by 0.4% (0.6%) compared to a point of average importance. At 30-40 (breakpoint) in the first game of the match, the probability of winning a point on service decreases by 0.8% (0.8%); at 5-5 in games and 30-40 in points in the first set, it decreases by 1.5% (2.1%); and at 5-5 in games and 30-40 in points in the fifth (third) set, it decreases by 4.6% (4.8%).

In order to test the iid hypothesis at point level and to analyze differences across players, we need point-to-point data over many matches involving different players. We were fortunate in obtaining point-to-point data on four years of Wimbledon men’s and women’s singles, 1992–1995, distributed over 481 matches, leading to 57,319 points in the men’s singles and 28,979 points in the women’s singles. This is one respect where our paper differs from the existing literature on the statistical analysis of tennis, which is hampered by a serious lack of detailed data. If some data are available, they are either based on end-of-match results (6-4, 6-3, 6-3 say) of several matches (Croucher 1981; Jackson and Mosurski 1997), or occasionally on a point-to-point analysis of

one match, usually an important final (Croucher 1995). The current paper is the first paper using data for many matches and at point level. The data concern only Wimbledon, one of the tournaments played on fast grass courts, and the generality of our conclusions may be restricted by this fact.

One characteristic of our data is that it involves heterogeneous players. If one estimates the probability of winning a point on service using pooled data without a proper correction for the quality of players, one will find that winning the previous point has a positive impact, even if points for each player individually are independent. The reason for this effect is that winning the previous point contains a small but positive piece of information about a player's quality. We call this 'pseudo-dependence', and it should be carefully distinguished from true dependence (the kind we are interested in): the presence of an effect of the past on the current point for one player.

In order to distinguish 'pseudo-dependence' from true dependence we correct for the quality of a player. But only part of the quality is observable (the player's ranking for example), while the rest is unobservable ('form of the day', fear against a specific opponent). This is our first problem. To solve this problem we model the unobserved part of quality as a random individual effect, in the same spirit as one typically corrects for unobserved heterogeneity in panel data (Hsiao 1986, p. 33). Hence, we explicitly use panel data techniques to correct for 'pseudo-dependence' and view our data set as a panel, that is a cross section of matches where each match consists of two time series of service points, one for each player.

Although panel data techniques are appropriate for our tennis data, direct application involves two additional problems. First, our focus variable (winning a point on service) is a binary variable—it can only take the values 0 (if the point is lost) and 1 (if the point is won)—while standard panel techniques are designed for continuous data. Secondly, most of our regressors are dynamic; for example, the performance on previous points, which captures dependence, is a function of lagged values of the dependent variable. The usual panel estimators, such as the within-group estimator, then become inconsistent (Hsiao 1986, p. 72).

The estimation of dynamic panel models with a discrete dependent variable is essentially an unsolved problem in the classical statistics literature, although there exist partial solutions, for example the logit approach by Honoré and Kyriazidou (2000). The Bayesian literature (Albert and Chib 1993; Chib and Greenberg 1994; Johnson and Albert 1999) provides also solutions for the probit and other link functions. Moreover, the Bayesian approach is exact rather than asymptotic. However, our very large data set makes the latter advantage irrelevant and, more importantly, it makes the computational burden in the Bayesian approach excessive.

To solve the estimation problem we exploit the special nature of our tennis data. This allows us to use the linear probability model to obtain results very similar to probit and logit (section 3). Thus we can use standard dynamic panel techniques (Hsiao 1986, chap. 4; Mátyás and Sevestre 1992, chap. 6; Baltagi 1995, chap. 8). We estimate our model by feasible GLS, which we show to be consistent for our dynamic panel, taking full account of the effects of the binary structure on the first two moments of the observations and the non-observables. Hence we provide a practical solution to the estimation of binary dynamic panels, and illustrate its effectiveness by applying it to our Wimbledon data.

In the statistical literature on tennis, only a small number of papers is concerned with some aspect of independence and identical distribution of points. Croucher (1981), studying the ‘back-to-the-wall effect’ (the possible effect that a player or a team plays better if trailing) in tennis, finds only very slight evidence of this effect. Another notable example is Jackson and Mosurski (1997), who investigate whether ‘getting slammed during your first set might affect your next’. In other words, they challenge the independence assumption and conclude that there is dependence, possibly caused by ‘psychological momentum’. Magnus and Klaassen (1999a–c) test 11 tennis hypotheses, many of them relating to the iid hypothesis. Most hypotheses are rejected.

The question of independence and identical distribution has been a hot topic in other sports, in particular basketball and baseball. Dependence between points is called ‘streaks’ in baseball. Lindsey (1961) finds that the distributions of runs scored in different half-innings in baseball are not homogeneous (the first and third half-innings have the highest expected number of runs). Nevertheless, Lindsey concludes that scores can be replicated assuming independence of runs scored in different innings. Simon (1971, 1977) notices that of the 31 World Series played since World War II until 1975, 18 have lasted seven games (the maximum). From this he concludes that there must be a ‘back-to-the-wall effect’ where the team that is behind performs better, thus challenging the independence assumption. Siwoff, Hirdt and Hirdt (1987, p. 97) find that the probability of hitting well in a game is independent of whether the hitter is on a streak or not. Albright (1993) too does not find convincing evidence of streaks. However, commenting on Albright’s paper, Stern and Morris (1993) and Albert (1993) suggest alternative approaches which might lead to a different conclusion. Stern (1995) suggests however that there might be streaks. The question remains unresolved.

In basketball, dependence between points is known as the ‘hot hand’. Research on the hot hand started with psychologists Gilovich, Vallone and Tversky (1985), who conclude that people believe in the hot hand (not that

a hot hand actually exists). Larkey, Smith and Kadane (1989) examine game-data for 18 NBA players, in particular Detroit’s Vinnie Johnson, who has a reputation of being one of the most streaky of shooters. From their descriptive analysis they conclude that Johnson is indeed a streaky shooter and hence that the hot hand exists. In contrast, Camerer (1989) shows that, although the basketball market believes in the hot hand, in fact there is no such thing in basketball shooting. He finds that bets placed on teams with a current winning streak are more likely to be *losers* than winners, and he concludes that teams on winning streaks are mistakenly believed to be ‘hot’. In a comment, Brown and Sauer (1993) disagree with Camerer and show that hot-hand effects are present. Clearly, this issue is unresolved as well.

This paper is organized as follows. In section 2 we briefly discuss the data and the representativeness of the sample. In section 3 we discuss the various problems that arise in modeling a binary dynamic panel data set with random effects, and show how this model can be consistently estimated (details in the Appendix). Section 3 (and the Appendix) contain the theoretical part of the paper and can be read without any knowledge of or interest in tennis. In section 4 we discuss the choice of variables for our tennis application, present our estimation results, and test the iid hypothesis. Extensive sensitivity analyses and diagnostics are provided in section 5. Section 6 concludes.

## 2 The data

Our data consist of 481 matches played at the Wimbledon championships during 1992–1995: 258 matches in the men’s singles and 223 matches in the women’s singles. In each match we know the two players and the complete sequence of points. Because of the special nature of points played in tie-breaks, these points have been excluded from the analysis in this paper. Since men play for three won sets and women for two, we have about twice as many points for the men (57,319) than for the women (28,979). The data are described in detail in Magnus and Klaassen (1999a).

We do not have data on all matches played during the four years. In fact, we only have the matches played on one of the five ‘show courts’ (Centre Court and Courts 1, 2, 13, and 14), since these are the only courts where the data collection has taken place. As a result we have data on almost half of the 1,016 matches played. Typically, matches involving the most important players are scheduled on the show courts, and this causes an under-representation in the data set of matches involving weaker players. All results have been corrected for this selection problem by weighting the matches by the inverses of the sampling percentages. The (admittedly imperfect)

weighting procedure is discussed in detail in Magnus and Klaassen (1999b). In fact, whether we weight or not, we still reject the iid hypothesis; see section 5, Q9.

The focus variable in the current paper is the probability of winning a point on service. Averaging over all players, this probability is estimated as 0.645 (with a standard error of 0.002) in the men’s singles and 0.559 (0.003) in the women’s singles, a large difference. This is one important difference between men’s singles and women’s singles, but there are other differences, reported in Magnus and Klaassen (1999a–c). As a result, we keep the analyses for the men’s singles and the women’s singles separate.

### 3 Dynamic binary panel data with random effects

In this section we develop the model and describe a consistent method of estimation, details of which can be found in the Appendix. The tennis aspect is minimal in this section, so that the results should be of more general interest.

We regard our data as a panel consisting of  $N$  matches (258 in the men’s singles, 223 in the women’s singles). We assume throughout that matches are independent, and we begin therefore by considering one match.

We wish to test the iid assumption and as a vehicle for doing so we model the probability of winning a point on service. Thus, let  $y_{at}$  be 1 if player  $\mathcal{A}$  wins his/her  $t$ -th service point (against player  $\mathcal{B}$ ) and 0 otherwise. (We write  $y_{at}$  instead of  $y_{abt}$  for simplicity of notation, although  $y_{at}$  depends on both  $\mathcal{A}$  and  $\mathcal{B}$ .) Similarly, let  $y_{bt}$  be 1 if  $\mathcal{B}$  wins his/her  $t$ -th service point (against  $\mathcal{A}$ ) and 0 otherwise.

Within each match of  $T$  points we have data on  $T_a$  service points of player  $\mathcal{A}$  and  $T_b$  service points of player  $\mathcal{B}$ . An average match in the men’s singles consists of  $T = T_a + T_b = 222$  points (130 points in the women’s singles).

The two players  $\mathcal{A}$  and  $\mathcal{B}$  in each match are modeled symmetrically. Concentrating on player  $\mathcal{A}$ , our starting point is the linear probability model

$$y_{at} = Q_a + D_{at} + \epsilon_{at}, \tag{1}$$

which consists of three components: quality  $Q_a$ , other (dynamic) regressors  $D_{at}$ , and random errors  $\epsilon_{at}$ . (Again, we write for instance  $Q_a$  instead of  $Q_{ab}$  for simplicity of notation.) Equation (1) says that the probability that  $\mathcal{A}$  wins the  $t$ -th service point is equal to the expectation of  $Q_a + D_{at}$  (assuming that  $\epsilon_{at}$  has expectation zero).

The choice of a linear probability model requires a justification. First, since  $E(y_{at}) = \Pr(y_{at} = 1)$  because of the binary character of  $y_{at}$ , the estimated expectation must lie between 0 and 1. This is not necessarily the case in general, but in our tennis data set it is always the case. In fact, the estimated probability of winning a point on service for each player lies in the interval (0.55, 0.75) in the men's singles and (0.42, 0.73) in the women's singles; see also section 5, Q1. Secondly, the binary character also implies second-moment restrictions on the model. The equality restriction  $E(y_{at}^2) = E(y_{at})$  will be imposed, while the second-moment inequality restrictions  $0 < E(y_{at}y_{as}) < 1$  and  $0 < E(y_{at}y_{bs}) < 1$  are satisfied (section 5, Q1). We don't consider higher-order moment restrictions, because our feasible GLS (FGLS) method of estimation only uses first and second moments. Thirdly, alternative non-linear models (such as probit and logit) will yield comparable results, because the relative deviation in the range (0.35, 0.70) between the link functions of the linear probability model and the probit and logit models (appropriately centered and scaled) is less than 1%; see Maddala (1983, p. 23) and Hsiao (1986, p. 155). Given that the linear and non-linear models will give similar results, we prefer the linear probability model since it is easier to use than panel logit or probit models and allows us to use standard (and computationally fast) GLS techniques in solving the dynamics problem (section 3.4).

Our approach to the problem is classical, not Bayesian, although Albert and Chib (1993) and Johnson and Albert (1999, chap. 3) show how to fit probit and other models using data augmentation and Gibbs sampling. In addition, Chib and Greenberg (1994) describe Bayesian fitting algorithms for ARMA models which can be extended, by data augmentation, to discrete response models. We prefer the classical approach for three reasons: our data set is large and hence the difference in results between the classical and Bayesian analysis will be small; the computational burden in using the Bayesian approach will be excessive in our case; and, since the linear probability model yields very similar results to probit and logit for the tennis data (see above), there is no reason to use the more demanding Bayesian methods needed to estimate logit or probit models.

We shall discuss each of the three components  $Q_a$ ,  $D_{at}$  and  $\epsilon_{at}$  in equation (1) in turn. The quality of player  $\mathcal{A}$  against player  $\mathcal{B}$ , denoted by  $Q_a$ , is a crucial ingredient, since it enables us to distinguish 'pseudo-dependence' from true dependence, as discussed in the introduction. Pseudo-dependence is closely related to the problem of spurious state dependence or heterogeneity bias in the panel data literature. The problem occurs when we ignore heterogeneous intercepts in a pooled regression, and it leads to biased slope estimates (see Hsiao 1986, p. 6). A commonly used method to avoid hetero-

geneity bias in panel models is to use individual-specific intercepts. In our context of modeling the probability of winning a service point in tennis,  $Q_a$  is such an individual intercept. Section 3.1 discusses quality in more detail.

While  $Q_a$  contains characteristics of  $\mathcal{A}$  and  $\mathcal{B}$  at the beginning of the match, the dynamic term  $D_{at}$  depends on all match information (of both players) up to but excluding point  $t$ . If points were iid, then such match information would be useless in predicting  $y_{at}$ . Hence, the variable  $D_{at}$  captures departures from the iid hypothesis, such as ‘dependence’ variables capturing a ‘winning mood’ (if such a mood exists), and characteristics of the point being played (such as the importance of a point). See section 3.2 for further details.

The random error  $\epsilon_{at}$  in (1) is affected by the binary structure of  $y_{at}$ , because it can only take the values  $0 - Q_a - D_{at}$  and  $1 - Q_a - D_{at}$ . The implications of the binary structure on the error term are discussed in section 3.3.

### 3.1 Quality

The proposed quality variable  $Q_a$  contains some components that we observe (most notably the ranking of the two players) and many that we do not observe (such as ‘form of the day’, fear against a specific opponent, and special ability, if any, on grass). We assume that observed quality is linear and denote it by  $x'_a\beta$ . (The specific ranking-based definition of  $x_a$  for our tennis data is discussed in section 4.1.) Unobserved quality is denoted by  $\eta_a$  and we model it as a random individual effect, just as one typically corrects for unobserved heterogeneity in the panel data literature.

Thus motivated we write quality as

$$Q_a = x'_a\beta + \eta_a. \quad (2)$$

We assume that the observed part contains a constant term, so that there is no loss in generality in assuming  $E(\eta_a) = E(\eta_b) = 0$ . In addition, we impose

$$\text{var}(\eta_a) = \text{var}(\eta_b) = \tau^2, \quad \text{cov}(\eta_a, \eta_b) = \gamma, \quad (3)$$

where  $|\gamma| < \tau^2$ . The assumption that the variance  $\tau^2$  of the random effect is constant across players, is standard (Hsiao 1986, p. 33), but the introduction of a covariance  $\gamma$  between the individual effects of both players in one match is not standard. This covariance captures the idea that if  $\mathcal{A}$  performs better on service than the rankings suggest, then one would expect that the probability that  $\mathcal{B}$  will win a point on service is lower. It also captures, for example, fear



against a specific opponent. Hence, for our tennis data we expect (but do not impose) that  $\eta_a$  and  $\eta_b$  are negatively correlated ( $\gamma < 0$ ).

The final assumption concerning quality is that the observed and unobserved parts are uncorrelated:

$$\text{cov}(x_a, \eta_a) = \text{cov}(x_a, \eta_b) = 0. \quad (4)$$

This is reasonable for our tennis application, since the rankings in  $x_a$  are determined well before the match starts (they are fixed at the end of the tournaments played in the week before Wimbledon). The assumption is also necessary, because otherwise the FGLS estimation procedure discussed in section 3.4 will not be consistent (Kiviet 1995).

### 3.2 Dynamics

In contrast to the component  $Q_a$ , which captures the effects of variables that are known before the match begins,  $D_{at}$  captures the effects of variables that change during the match. We write

$$D_{at} = z'_{at}\delta \quad (5)$$

and interpret  $z_{at}$  as variables which reflect deviations from the iid assumption. In particular, previous points may influence the current point. This is *dependence* and the variable  $y_{a,t-1}$  is an obvious (be it simple) example. In addition, the current point may be played differently from other points. If this occurs we have *non-identical distribution*, which could be measured by the importance of point  $t$ . In section 4.2 we shall specify the dynamic regressors in such a way that deviations from the iid assumption (if present) are allowed to be heterogeneous across players.

In our application, the regressors  $z_{at}$  are completely determined by the history of the match up to point  $t$  and possibly exogenous attributes (such as the ranking of both players). We emphasize that the development of the match depends also on  $\eta_a$  and  $\eta_b$ , so that  $\text{cov}(z_{at}, \eta_a)$  and  $\text{cov}(z_{at}, \eta_b)$  may be nonzero.

### 3.3 Error term: the effect of binary structure

The third component in (1) is the error term  $\epsilon_{at}$ . The binary structure of  $y_{at}$  has implications for the error term  $\epsilon_{at}$  and in particular for its variance.

We assume that  $E(\epsilon_{at}) = 0$ . Regarding the second moments we assume that

$$\begin{aligned} \text{cov}(\epsilon_{at}, x_a) &= \text{cov}(\epsilon_{at}, x_b) = 0, & \text{cov}(\epsilon_{at}, \eta_a) &= \text{cov}(\epsilon_{at}, \eta_b) = 0, \\ \text{cov}(\epsilon_{at}, z_{as}) &= 0 \quad (s \leq t), & \text{cov}(\epsilon_{at}, z_{bs}) &= 0 \quad (s \leq S_{bt}), \\ \text{cov}(\epsilon_{at}, \epsilon_{as}) &= 0 \quad (s \neq t), & \text{cov}(\epsilon_{at}, \epsilon_{bs}) &= 0, \end{aligned} \quad (6)$$

where  $S_{bt}$  denotes the total number of points served by  $\mathcal{B}$  until the beginning of the current game (where  $\mathcal{A}$  is serving). These are standard assumptions in the panel data literature. For our tennis data set they are also reasonable, because the quality variables  $x_a$  and  $\eta_a$  are given at the beginning of the match and the dynamic regressors  $z_{at}$  only depend on the outcomes of the previous points.

In the panel data literature it is usually assumed that the variance of  $\epsilon_{at}$  is homoskedastic (Hsiao 1986, p. 33). In our case this is not possible, because of the binary character of the observations. Since  $E(y_{at}) = E(y_{at}^2)$ , we obtain

$$\begin{aligned} \text{var}(\epsilon_{at}) &= E\{(x'_a\beta + z'_{at}\delta)(1 - x'_a\beta - z'_{at}\delta)\} \\ &\quad - 2E\{(z'_{at}\delta)(\eta_a + \epsilon_{at})\} - \tau^2 \equiv \sigma_a^2, \end{aligned} \quad (7)$$

so that  $\text{var}(\epsilon_{at})$  depends on  $a$ . Hence we must take proper account of heteroskedasticity.

### 3.4 Estimation

Assumptions (1) to (7) imply the following dynamic binary panel model with random effects:

$$y_{at} = x'_a\beta + z'_{at}\delta + u_{at}, \quad u_{at} = \eta_a + \epsilon_{at}, \quad (8)$$

and similarly for player  $\mathcal{B}$ . Stacking the  $\{u_{at}\}$  into  $T_a \times 1$  vectors  $u_a$ , and defining  $\iota_a$  as the  $T_a \times 1$  vector of ones and  $I_{T_a}$  as the  $T_a \times T_a$  identity matrix, the  $T \times T$  variance matrix of the error vector  $(u'_a, u'_b)'$  of the whole match is given by

$$\Omega = \text{var} \begin{pmatrix} u_a \\ u_b \end{pmatrix} = \begin{pmatrix} \sigma_a^2 I_{T_a} + \tau^2 \iota_a \iota'_a & \gamma \iota_a \iota'_b \\ \gamma \iota_b \iota'_a & \sigma_b^2 I_{T_b} + \tau^2 \iota_b \iota'_b \end{pmatrix}. \quad (9)$$

The matrix  $\Omega$  is a generalization of the variance matrix of the standard two-error-components model, and plays a crucial role in the FGLS estimation procedure discussed below. (Again, we simplify notation by writing  $\Omega$  rather than  $\Omega_{ab}$ , although  $\Omega$  is different in each match.)

We wish to estimate the parameters of this model consistently. In dealing with consistency we shall always make the usual assumption that  $T_a$  (the number of points served by a player) is finite and that  $N$  (the number of matches) can become large. One could argue that  $T_a$  is large for our tennis data (see beginning of section 3) and that therefore the asymptotic approximation when  $T_a \rightarrow \infty$  will work satisfactorily. This, however, is not the case, as we shall see in section 5, Q8.

Estimation of a dynamic panel with random effects is no trivial exercise, mainly because  $z_{at}$  will be correlated with  $u_{at}$ , since variables such as  $y_{a,t-1}$  in  $z_{at}$  will be correlated with  $\eta_a$  in  $u_{at}$ . Hence the regressors and the error term are contemporaneously correlated, implying that OLS is inconsistent (even for large  $T_a$ ).

The within-group or least-squares dummy variables (LSDV) estimator does take account of  $\eta$ , but is also not appropriate in this dynamic context. If we average over points for each player and subtract from equation (8), we obtain  $y_{at} - \bar{y}_a = (z_{at} - \bar{z}_a)' \delta + (\epsilon_{at} - \bar{\epsilon}_a)$ . We have lost  $\eta_a$ , but again there is contemporaneous correlation and hence inconsistency (Nickell 1981; Kiviet 1995).

In the Appendix we describe how the parameters of the model can be consistently estimated by FGLS. We use FGLS rather than GMM (Arellano and Bover 1995; Ahn and Schmidt 1997), because there is much evidence that FGLS works well in finite samples (Balestra and Nerlove 1966; Sevestre and Trognon 1985), while Kiviet (1995) finds mixed results for the finite sample behavior of GMM in such models.

## 4 Application to the Wimbledon data

### 4.1 Specification of quality variables $x_a$

The ‘quality’ variables  $x_a$  should reflect the observed quality of player  $\mathcal{A}$  versus player  $\mathcal{B}$ . We base the definition of observed quality on the ranking of both  $\mathcal{A}$  and  $\mathcal{B}$  according to the lists published just before Wimbledon by the Association of Tennis Professionals (for the men) or the Women’s Tennis Association (for the women). These two lists contain the official rankings based on performances over the last 52 weeks, including last year’s Wimbledon. The ranking of player  $\mathcal{A}$  is denoted  $\text{RANK}_a$ . Note that  $\text{RANK}_a$  can be 500 even though only 128 players take part in the tournament.

Direct use of the rankings is unsatisfactory, because quality in tennis is a pyramid: the difference between the top two players (ranked 1 and 2) is generally larger than between two players ranked 101 and 102. The pyramidal

structure is also evident in the seeding system and plays a role in acquiring points for the official ranking lists. The pyramid is based on ‘expected round’: 8 for the player who is expected to win the final (round 7), 3 for a player who is expected to lose in round 3, *et cetera*. A problem with ‘expected round’ is that it does not distinguish, for example, between players seeded 9–16 since all of them are expected to lose in round 4.

Thus motivated, we propose a smoother measure of ‘expected round’ by transforming the ranking of each player into a variable  $R$  as follows:

$$R_a = 8 - \log_2(\text{RANK}_a), \quad (10)$$

where  $\log_2 x$  denotes  $\log x$  with base 2. For example, if  $\text{RANK} = 3$  then  $R = 6.42$ , while if  $\text{RANK} = 4$  then  $R = 6.00$ . Notice that players with  $\text{RANK} > 128$  are not expected to play at Wimbledon at all; for these players  $R < 1$ . Also,  $R$  can be negative, but this causes no problems. The average value of  $R$  over all players is 2.57 (men) and 2.74 (women). We shall show the robustness of our test results with respect to definition (10) in section 5, Q2.

What then should be included—apart from an intercept—in  $x_a$ ? Obviously, a player scores more points on service against a weaker than against a stronger opponent. Hence relative quality (‘gap between the two players’)  $R_a - R_b$  matters. But also absolute quality (‘overall quality of the match’)  $R_a + R_b$  may matter, since we know that in a match between two strong players more points on service are scored than in a match between two weaker players (Magnus and Klaassen 1999a). We thus write

$$x'_a = (1, (R_a - R_b), (R_a + R_b)), \quad (11)$$

and we let  $\beta = (\beta_0, \beta_1, \beta_2)'$  denote the corresponding vector of three unknown parameters. Finally we center both  $R_a - R_b$  and  $R_a + R_b$  by subtracting their sample means. This makes the interpretation of  $\beta_0$  more transparent: it gives the observed quality for an average match.

## 4.2 Specification of dynamic variables $z_{at}$

Since the focus of this paper is to test the iid hypothesis, the dynamic variables should contain ‘dependence’ variables and ‘non-identical distribution’ variables. In essence we capture the ‘dependence’ aspect by the previous point  $y_{a,t-1}$  and the ‘non-identical distribution’ aspect by the importance of the point. There are, however, three subtleties.

The first subtlety concerns the influence of the previous point. The point preceding the first point of a game is the final point of the previous service

game of the current server. Hence, in measuring the influence of the previous point, there is a difference between the first point (where the previous point is ‘long’ ago) and other points in a game. We thus define a dummy variable  $d_{at}$  which takes the value 1 if the current point  $t$  is the first point in a game, and 0 otherwise. We then define

$$\tilde{y}_{a,t-1} = \begin{cases} 0 & \text{if } d_{at} = 1, \\ y_{a,t-1} & \text{if } d_{at} = 0. \end{cases} \quad (12)$$

Using both  $\tilde{y}_{a,t-1}$  and  $d_{at}$  allows us to study the effect of the previous point without having to delete the first point in every game.

The second subtlety is the measurement of the importance of a point. The importance of the  $t$ -th point served by  $\mathcal{A}$ , denoted  $imp_{at}$ , is defined as the probability that  $\mathcal{A}$  will win the match if he/she wins the current point minus the probability that  $\mathcal{A}$  will win the match if he/she loses the current point. This definition was first suggested by Morris (1977).

In the calculation of the importance  $imp_{at}$ , we assume that associated with each match there are two pre-match (fixed) probabilities  $p_a$  and  $p_b$ . We define  $p_a$  as the estimated pre-match probability that  $\mathcal{A}$  wins a point on service. In particular,  $p_a \equiv x'_a \tilde{\beta}$ , where  $\tilde{\beta}$  is the estimate of  $\beta$  under the assumption that  $\delta = 0$ . We treat  $p_a$  as a constant, not an estimate. The pre-match probability  $p_b$  is similarly defined. Given the structure of a Wimbledon match (best-of-three sets for women, best-of-five sets for men; tie-break at 6–6, no tie-break in deciding set), given the two fixed pre-match probabilities, and assuming that the points are independent and identically distributed, our computer program calculates exactly the probability that  $\mathcal{A}$  wins the match at each point of the match, and hence  $imp_{at}$  can be calculated at each point. The average of  $imp_{at}$  over all points and players is 0.028 (men) and 0.034 (women). The distribution of  $imp_{at}$  is skewed to the right, indicating that important points are rare.

There is an apparent contradiction in the fact that, in calculating the importance of a point, we assume that points are iid ( $\delta = 0$ ), while this is just what we intend to test. As a result, importance could be misrepresented. The question here is not, however, whether points are iid but what the *effect* is of the iid assumption on the importance measure. The procedure is justified since  $p_a$  provides a very good first-order approximation to the probability that  $\mathcal{A}$  wins a service point, as the deviation from iid is small (section 4.3). Having defined the three explanatory variables, we now write the dynamic part as  $D_{at} = \delta_1 \tilde{y}_{a,t-1} + \delta_2 d_{at} + \delta_3 imp_{at}$ .

There is, however, a third subtlety to be taken into account. It may be the case that possible deviations from the iid assumption are player-dependent.

For example, top-players may be more equable (more iid) than lesser players. We account for this type of heterogeneity by letting  $\delta_i$  depend on the rankings:

$$\delta_i = \delta_{i0} + \delta_{i1}(R_a - R_b) + \delta_{i2}(R_a + R_b) \quad (i = 1, 2, 3). \quad (13)$$

Since  $(R_a - R_b)$  and  $(R_a + R_b)$  are centered (section 4.1),  $\delta_{i0}$  gives the effect for an average match.

This then leads to the following definition of the dynamic regressors,

$$z'_{at} = (\tilde{y}_{a,t-1}x'_a, d_{at}x'_a, imp_{at}x'_a), \quad (14)$$

and the associated parameters  $\delta' = (\delta_{10}, \delta_{11}, \delta_{12}, \delta_{20}, \delta_{21}, \delta_{22}, \delta_{30}, \delta_{31}, \delta_{32})$ . If  $\delta_{10} > 0$  then there is positive dependence ('winning mood'), so that winning the previous point increases the probability of winning the current point. This dependence is smaller for players who are better than their opponent (if  $\delta_{11} < 0$ ) and in matches between two good players (if  $\delta_{12} < 0$ ). If  $\delta_{30} < 0$  then it is more difficult for the server to win an important point. The better player in a match can somewhat neutralize this effect if  $\delta_{31} > 0$ . If  $\delta_{32} > 0$ , then in matches between two good players the effect is also smaller.

Our main interest is the iid hypothesis, which now takes the form  $\delta = 0$ . Also of interest is the 'homogeneity' hypothesis, which tests whether deviations from iid (if present) are homogeneous among players. The second hypothesis takes the form  $\delta_{i1} = \delta_{i2} = 0 \quad (i = 1, 2, 3)$ .

### 4.3 Estimation results and test of the iid hypothesis

We estimate the fourteen unknown parameters (three  $\beta$ 's, nine  $\delta$ 's,  $\tau^2$  and  $\gamma$ ) of our model by FGLS as explained in section 3.4, based on the Wimbledon data described in section 2. The results (not reported) show that four of the nine  $\delta$ -parameters (the same four for men and women) are not significantly different from zero ('significant' is always at the 5% level), both individually and simultaneously (Wald tests yield  $p$ -values of 96% (men) and 19% (women)). The non-significant parameters are  $\delta_{11}$  (previous point  $\times$  quality difference) and all three  $\delta$ -parameters associated with the first-point-in-game dummy  $d_{at}$ . We delete these four  $\delta$ -parameters and obtain a reduced model containing ten unknown parameters. This model forms the basis for all subsequent discussion. The estimates of the reduced model are presented in Table 1 (standard errors in parentheses,  $p$ -values in square brackets).

Table 1. Estimation results for the Wimbledon Data

|                                                      | Men's singles |          | Women's singles |          |
|------------------------------------------------------|---------------|----------|-----------------|----------|
| constant ( $\beta_0$ )                               | 0.6456        | (0.0040) | 0.5596          | (0.0050) |
| quality difference ( $\beta_1$ )                     | 0.0106        | (0.0014) | 0.0198          | (0.0017) |
| quality sum ( $\beta_2$ )                            | 0.0035        | (0.0014) | 0.0040          | (0.0017) |
| <i>previous point <math>\tilde{y}_{a,t-1}</math></i> |               |          |                 |          |
| × constant ( $\delta_{10}$ )                         | 0.0085        | (0.0041) | 0.0123          | (0.0058) |
| × quality sum ( $\delta_{12}$ )                      | −0.0028       | (0.0014) | −0.0052         | (0.0019) |
| <i>importance <math>imp_{at}</math></i>              |               |          |                 |          |
| × constant ( $\delta_{30}$ )                         | −0.1304       | (0.0666) | −0.1752         | (0.0779) |
| × quality difference ( $\delta_{31}$ )               | 0.0394        | (0.0262) | 0.1405          | (0.0493) |
| × quality sum ( $\delta_{32}$ )                      | 0.0533        | (0.0231) | 0.0137          | (0.0241) |
| <i>random effects</i>                                |               |          |                 |          |
| variance ( $\tau^2$ )                                | 0.0031        |          | 0.0026          |          |
| correlation ( $\gamma/\tau^2$ )                      | −0.5353       |          | −0.8317         |          |
| <i>Wald tests</i>                                    |               |          |                 |          |
| iid                                                  | 22.9901       | [0.0003] | 25.2179         | [0.0001] |
| homogeneity                                          | 14.6005       | [0.0022] | 16.1926         | [0.0010] |

The main conclusion from Table 1 is that the iid hypothesis ( $\delta_{10} = \delta_{12} = \delta_{30} = \delta_{31} = \delta_{32} = 0$ ) is strongly rejected with  $p$ -values of 0.03% (men) and 0.01% (women). (In the unrestricted model the iid test is also rejected with  $p$ -values of 0.50% (men) and 0.03% (women).) The next section shows that this rejection is robust. The lack of independence and the lack of identical distribution are about equally important in this rejection. The independence hypothesis ( $\delta_{10} = \delta_{12} = 0$ ) is rejected with a  $p$ -value of 1.7% (men) and 0.3% (women), while the identical distribution hypothesis ( $\delta_{30} = \delta_{31} = \delta_{32} = 0$ ) is rejected with a  $p$ -value of 1.7% (men) and 0.5% (women).

Player homogeneity ( $\delta_{12} = \delta_{31} = \delta_{32} = 0$ ) is also rejected with  $p$ -values of 0.2% (men) and 0.1% (women), respectively. (In the unrestricted model the homogeneity test is rejected with  $p$ -values of 1.9% (men) and 0.15% (women).) Hence, deviations from iid are player-dependent.

Winning the previous point has a positive effect on winning the current point, both for men and for women, since  $\delta_{10} > 0$ . The stronger the players, the weaker is this effect:  $\delta_{12} < 0$ . At important points the server has a disadvantage:  $\delta_{30} < 0$ . The effect is weaker for high-level matches than for low-level matches ( $\delta_{32} > 0$ , although not significantly in the women's singles) or when the server is better than the receiver ( $\delta_{31} > 0$ , although not significantly for the men). Observed quality matters. As expected, the relative quality of the two players ('quality difference') is more important than the absolute quality ('quality sum'):  $\beta_1 > \beta_2$ . Comparing men's and

women’s singles, equality of the  $\beta$ -parameters is obviously rejected (mainly because  $\beta_0$  is very different for men and women). We also see that  $\beta_1$  is larger for the women than for the men, which corresponds to the fact that the difference in strength between top players and lesser players is larger in the women’s singles than in the men’s singles.

Even though the iid hypothesis is rejected, one might have expected a stronger rejection, since the sample is so large. Although seven of the ten  $\delta$ -estimates are significant, their  $t$ -ratios are between 2 and 3, and hence individually only marginally significant. Maybe iid is rejected, not because the null hypothesis is false, but because the standard errors are so small, due to the large sample. To examine this possibility, we need a test for iid that does not depend on the standard errors. We thus consider the *signs* of the  $\delta$ -estimates (Berkson 1938) and see that all five estimates have the same sign for men and women (Table 1). Since the men and women samples are independent, this has probability  $1/32$  if points are iid (assuming for simplicity that the elements of  $\hat{\delta}$  are mutually uncorrelated for both men and women; taking proper account of the correlation estimates yields a probability of 3.3%, very close to  $1/32$ ). Hence, without using the effect of the large sample on the standard errors, we reject iid as well.

We also compare the *magnitudes* of the  $\delta$ -estimates between men and women. Equality of the  $\delta$ -parameters for these two independent samples is *not* rejected ( $p$ -value equals 33.1%). This shows that, even with our large sample, not every hypothesis is rejected, and that there is considerable consistency of the results between men’s singles and women’s singles. Both findings support our conclusion that points in tennis are not iid.

Our results also suggest that—even though iid is rejected—the assumption of iid in specific applications (such as forecasting) could be relatively harmless. The results of the next section support this statement.

## 5 Diagnostics and sensitivity analysis

Clearly our model—like all models—is imperfect. In this section we investigate whether the underlying assumptions are justified and how sensitive the main focus of the paper (testing the iid hypothesis) is to possible imperfections of the model. The first activity is called diagnostic testing, the second sensitivity analysis. We ask and answer eleven questions, some (or all) of which may already have occurred to the reader.

*Q1: Is the use of the linear probability model justified?* We have already commented—in section 3, below (1)—on the differences between logit, probit, and the linear probability model and argued that the linear probability model



is appropriate in our case. The main reason why the linear probability model could fail is if estimated probabilities fall outside the (0,1) interval. This is never the case. We already know that the estimate of  $E(y_{at}) = \Pr(y_{at} = 1)$  lies in the interval (0.55, 0.75) for men and (0.42, 0.73) for women. Moreover, at each point,  $\hat{y}_{at}$  lies in the interval (0.37, 0.91) for men and (0.32, 0.85) for women. Further, the estimate of  $E(y_{at}y_{as})$  lies in the interval (0.31, 0.57) for men and (0.18, 0.53) for women for all  $0 < |t - s| \leq 10$ . For  $|t - s| > 10$  not much will change, since the boundaries are very stable for different  $t - s$ . Finally, the estimated  $E(y_{at}y_{bs})$  also falls in the (0,1) range for all players.

*Q2: Is our basic quality measure appropriate?* We introduced our basic quality measure in (10):  $R_a = 8 - \log_2(\text{RANK}_a)$ . We defended this choice on theoretical grounds. If, in contrast, we assume a linear basic quality measure  $R_a = -\text{RANK}_a$ , then the iid-hypothesis is also rejected with  $p$ -values of 0.17% (men) and 0.07% (women).

*Q3: Is unobserved quality relevant?* Put differently, are random effects actually present, which translates to the null hypothesis  $\eta_a = \eta_b = 0$ ? We use a Hausman test based on our FGLS estimator and the restricted FGLS estimator of  $\beta$  and  $\delta$ . (The restriction is  $\eta_a = \eta_b = 0$  implying a diagonal  $\Omega$  matrix.) Under the null hypothesis, the restricted FGLS estimator is consistent and efficient. Under the alternative, FGLS is consistent and restricted FGLS is inconsistent. The usual Hausman test, based on the difference between the two estimators and the difference between the two variance matrices, has asymptotically a  $\chi^2(8)$  distribution and takes the values 81.8 (men) and 45.5 (women) with  $p$ -values of less than 0.01%. The null hypothesis is thus very firmly rejected, unobserved quality is definitely relevant, and random effects should play a role in the estimation procedure.

*Q4: Is there any autocorrelation left in the residuals?* We consider the FGLS residuals weighted by  $\hat{\Omega}^{-1/2}$  and denote these by  $\hat{v}_t$ . We regress  $\hat{v}_t$  on  $\hat{v}_{t-k}$  for  $k = 1, 2, \dots$  and this gives estimates of the correlations  $r_k$  (with standard errors). Since each game contains at least four points, it is natural to consider only  $r_1, r_2$  and  $r_3$ . These are all insignificant for both men and women. Moreover, a joint Wald test of  $r_1 = r_2 = r_3 = 0$  gives  $p$ -values of 62.4% (men) and 28.2% (women). Hence, there is no evidence of residual autocorrelation.

*Q5: Are the non-iid effects appropriately measured?* We use  $y_{a,t-1}$  to test independence, and importance  $imp_{at}$  to test whether points are identically distributed. We could replace  $y_{a,t-1}$  by some other measure of dependence such as ‘outperformance’ of the server in the current game with respect to the current set—defined as relative frequency of points won in current game minus relative frequency of points won in previous service games of current

set. (Albright (1993), Albert (1993), and Stern and Morris (1993) discuss similar measures in baseball.) Similarly, we could replace  $imp_{at}$  with a break-point dummy. The iid hypothesis is rejected in both cases.

*Q6: Are not more regressors needed?* We have experimented *inter alia* by adding  $y_{a,t-2}$  with an associated dummy, various measures of ‘outperformance’ of the server in the current game with respect to the current set, various measures of ‘outperformance’ of the server in the current set with respect to previous sets, similar measures for the receiver in his/her service games, a breakpoint dummy, and separation of the importance of point-in-match ( $imp_{at}$ ) into its three components: importance of point-in-game, game-in-set, and set-in-match (Morris 1977). Although occasionally the extra parameters are significant, the iid hypothesis is rejected in each of the extended models, showing the robustness of our result.

*Q7: Should the model allow for non-linearities?* We already allow for one form of non-linearity by making  $\delta$  dependent on the quality of the player and his/her opponent, see (13). But we have also experimented by adding  $y_{a,t-1} \times imp_{at}$  and  $imp_{at}^2$ . The coefficients of these two non-linear terms are not jointly significant ( $p$ -values of 86.0% (men) and 22.5% (women)) and inclusion of the extra terms does not alter the outcome of the iid test.

*Q8: Are the results sensitive to the method of estimation?* Our method of estimation is FGLS (section 3.4). But there are other estimation methods. Anderson-Hsiao’s method does *not* reject the iid hypothesis ( $p$ -values are 24.1% (men) and 17.1% (women)), although the resulting estimates do not contradict our FGLS estimates. Both findings are not surprising, because it is well-known that Anderson-Hsiao, though consistent, is not efficient. Hence, when effects are small like in our tennis hypothesis, Anderson-Hsiao may not detect them. This motivates our desire for more efficient methods such as FGLS. One could argue that  $T_a$  is large for our tennis data and that therefore the asymptotic approximation when  $T_a \rightarrow \infty$  will work satisfactorily (see section 3.4). For  $T_a \rightarrow \infty$  the within-group estimator (also known as the LSDV estimator) is consistent (Hsiao 1986, p. 74). However, use of this estimator leads to an estimate of  $\delta_{10}$  which is far from significant. Hence, although  $T_a$  is ‘large’, it is not large enough to justify an asymptotic approximation for  $T_a \rightarrow \infty$ . Nevertheless, iid is still clearly rejected.

*Q9: What happens if we do not weight the data?* We explained in section 2 that the data are weighted to account for the under-representation of weaker players in the sample. If we do not weight the data, then the iid test still rejects with  $p$ -values of 0.6% (men) and 0.9% (women).

Table 2. In-Sample and Out-of-Sample Fit

| Model    | In-sample     |        |                 |        | Out-of-sample |        |                 |        |
|----------|---------------|--------|-----------------|--------|---------------|--------|-----------------|--------|
|          | Men's singles |        | Women's singles |        | Men's singles |        | Women's singles |        |
|          | RMSE          | MAE    | RMSE            | MAE    | RMSE          | MAE    | RMSE            | MAE    |
| non-iid  | 0.4777        | 0.4557 | 0.4936          | 0.4865 | 0.4778        | 0.4558 | 0.4938          | 0.4866 |
| iid      | 0.4777        | 0.4559 | 0.4936          | 0.4869 | 0.4778        | 0.4559 | 0.4937          | 0.4869 |
| constant | 0.4787        | 0.4577 | 0.4965          | 0.4928 | 0.4787        | 0.4577 | 0.4965          | 0.4928 |

*Q10: How is the in-sample fit?* The usual  $R^2$  is misleading as a diagnostic, since we are dealing with binary variables (Maddala 1983, p. 38). Instead, we use the root mean squared error (RMSE) and the mean absolute error (MAE) as our measures of fit, and we compare our model with the ‘iid model’ (where all  $\delta$ ’s are zero). Our model (the ‘non-iid’ model) appears to perform only marginally better (lower MAE) than the ‘iid model’, see the left panel of Table 2. As a reference point we also estimate the ‘constant model’ (where also the  $\beta$ ’s are zero except the constant). Our model and the ‘iid model’ have a much better fit than the ‘constant model’. This means that including the ranking variables improves the fit substantially, while including the non-iid variables does so only marginally.

*Q11: How is the out-of-sample fit?* We analyze the out-of-sample fit by estimating our model based on three years and then predicting the fourth. We can take the first three years and predict the fourth year, but we can also, for example, estimate the last three years and predict the first. In this way we obtain four predictions. To simplify the presentation we average across years. The conclusions are the same as for the in-sample analysis (see the right panel of Table 2). It also appears that the role of outliers is very limited—these would yield a good in-sample fit, but a bad out-of-sample fit. This is another proof of the robustness of our results. Hence, our non-iid model survives out-of-sample analysis. In fact, all diagnostics and sensitivity analyses show that the rejection of iid based on the estimated model is robust in many directions against deviations from the underlying assumptions.

## 6 Concluding remarks

In this paper we use 86,298 points (481 matches) at Wimbledon 1992–1995 to investigate whether points in professional tennis are independent or identically distributed. We reject this hypothesis and the rejection is robust. Winning the previous point has a positive effect on winning the current point and at important points the server has a disadvantage. The paper also

shows that the deviations from iid depend on the quality of the players: the stronger a player, the smaller the deviation from the iid hypothesis. These results are the same for men and women. This suggests that players should be trained to ‘play every point as it comes’.

Even though we show that points are not iid, we also show that the divergence from iid is small. Hence, in many practical applications concerning tennis—such as predicting the winner of the match while the match is in progress—the iid hypothesis will still provide a good approximation, *if we correct for the quality of the players*. This makes our study a useful starting point for future papers on tennis. Both the large data set and the accurate FGLS method of estimation are essential in detecting the small deviation from iid.

In addition to the empirical findings on tennis, the paper provides a theoretical contribution to the estimation of discrete dynamic panel data models. We provide a structure which allows for a binary dependent variable, dynamic regressors, and random individual effects, and is nevertheless easy and fast to estimate by feasible GLS. Since estimation of binary dynamic panels is still an unsolved problem in the classical statistical literature, this approach may find application outside tennis as well.

## Appendix: FGLS estimation

FGLS estimation consists of two stages: consistent estimation of (the parameters in)  $\Omega$  given in (9), and GLS on (8) using the estimated  $\Omega$ .

*Consistent estimation of  $\Omega$ .* In order to estimate  $\Omega$  consistently we need consistent estimates of  $\beta$ ,  $\delta$ ,  $\tau^2$  and  $\gamma$  (see (9) and (7)). These are obtained in three steps. In step 1 we take first differences in (8) to remove the random effect  $\eta_a$ ,

$$y_{at} - y_{a,t-1} = (z'_{at} - z'_{a,t-1})\delta + (\epsilon_{at} - \epsilon_{a,t-1}), \quad (\text{A.1})$$

and estimate  $\delta$  from this equation. Since  $z_{at}$  is correlated with  $\epsilon_{a,t-1}$ , we need to estimate  $\delta$  by instrumental variables. We follow Anderson and Hsiao (1981) by using  $z_{a,t-1}$  as an instrument for  $z_{at} - z_{a,t-1}$ , because it is uncorrelated with the error term and correlated with the explanatory variable (Arellano 1989; Kiviet 1995; Judson and Owen 1999). We thus obtain a consistent estimate  $\hat{\delta}$  of  $\delta$ . In step 2 we use  $\hat{\delta}$  to rewrite equation (8) as

$$y_{at}^* = x_a' \beta + u_{at}^*, \quad y_{at}^* = y_{at} - z_{at}' \hat{\delta}, \quad u_{at}^* = u_{at} - z_{at}' (\hat{\delta} - \delta). \quad (\text{A.2})$$

Regressing  $y_{at}^*$  on  $x_a$  in (A.2) yields a consistent estimate  $\hat{\beta}$  of  $\beta$  (Anderson and Hsiao 1982). Step 3 is motivated by Hsiao (1986, p. 89). We use the consistent estimates  $\hat{\delta}$  and  $\hat{\beta}$  to estimate the covariance parameters  $\tau^2$  and  $\gamma$ . Since  $u_{at} = \eta_a + \epsilon_{at}$  and letting  $\bar{u}_a = (1/T_a) \sum u_{at}$ , we find that

$$E(\bar{u}_a^2) = \tau^2(1 - \frac{1}{T_a}) + \frac{1}{T_a}(\tau^2 + \sigma_a^2), \quad \gamma = E(\bar{u}_a \bar{u}_b), \quad (\text{A.3})$$

where  $\sigma_a^2$  is restricted by (7). We replace the errors  $u_{at}$  by the residuals  $\hat{u}_{at} = y_{at} - x_a' \hat{\beta} - z_{at}' \hat{\delta}$  and estimate  $\tau^2 + \sigma_a^2$  by replacing  $\beta, \delta$  and  $\eta_a + \epsilon_{at}$  ( $= u_{at}$ ) in (7) by their estimates and the expectations by the appropriate sample means. Averaging (A.3) over players then yields consistent estimates  $\hat{\tau}^2$  and  $\hat{\gamma}$ . Subtracting  $\hat{\tau}^2$  from the estimated  $\tau^2 + \sigma_a^2$  yields  $\hat{\sigma}_a^2$  for all players. We thus obtain the consistent  $T \times T$  variance matrix  $\hat{\Omega}$  from (9).

*Consistency of GLS.* We next show that GLS (with known  $\Omega$ ) is consistent. Since OLS is inconsistent, the consistency of GLS is not trivial. It rests on two key ingredients: the absence of initial conditions and the fact that observed quality is not correlated with the random effect; see (4).

Regarding the first ingredient, it is typical in economic data sets that for small  $t$  the dynamic regressors  $z_{at}$  depend on endogenous variables from

before the sample period, such as  $y_{a0}$ . If such initial values are correlated with  $\eta_a$ , GLS becomes inconsistent (Hsiao 1986, p. 88). Our case is different, because the process that generates the data coincides with the start of the actual data collection, so that  $z_{at}$  can not depend on variables such as  $y_{a0}$ —there are no points before the beginning of the match. Hence, inconsistency due to the existence of initial conditions does not occur here (Sevestre and Trognon 1985; Nerlove and Balestra 1992; Kiviet 1995; Blundell and Bond 1998).

If both ingredients are present, then Anderson and Hsiao (1982) and Hsiao (1986, p. 88) show—under normality—that GLS is consistent since GLS then equals maximum likelihood. Because of the binary structure of  $y_{at}$ , we do not assume normality. But it is obvious in our case that GLS equals normality-based pseudo maximum likelihood. Gouriéroux, Monfort and Trognon (1984) show that the pseudo maximum likelihood estimator is consistent. Hence GLS is consistent too.

Now that we know that  $\hat{\Omega}$  is consistent and that GLS is consistent, we obtain new consistent estimates of  $\beta$  and  $\delta$  by performing GLS on (8), using  $\hat{\Omega}$  instead of  $\Omega$ . These FGLS estimates will be more efficient than the ones obtained from (A.1) and (A.2). The efficiency gain is important, see section 5, Q8.

As a refinement of the procedure we use the FGLS estimates of  $\beta$  and  $\delta$  to form a new estimate of  $\Omega$  and continue this process until convergence. The estimation results are *not* sensitive to this iterative procedure.

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